

OCR Further Mathematics
Pure Core - Further Vectors
Mark Scheme
AS and A Level
Further Mathematics A - H235, H245



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The latest topic questions for the year 2024 / 2025+ exams.
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Question			Answer/Indicative content	Marks	Part marks and guidance	
1	a		$\vec{AB} = \begin{pmatrix} -3 \\ 3 \\ 3 \end{pmatrix} \text{ oe}$ Equation of AB is $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 4 \\ 2 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} -3 \\ 3 \\ 3 \end{pmatrix} \text{ oe}$ $\Rightarrow 4 - x = y - 2 = z$	B1 (AO 1.1) M1 (AO 1.1) A1 (AO 1.1) [3]	soi their $\begin{pmatrix} -3 \\ 3 \\ 3 \end{pmatrix}$ soi r = or $\begin{pmatrix} x \\ y \\ z \end{pmatrix} =$ is not required for M1 Allow equivalent equations e.g. $\Rightarrow 1 - x = y - 5 = z - 3$ from using B	
	b		$\vec{AB} = \begin{pmatrix} -3 \\ 3 \\ 3 \end{pmatrix}, \vec{OM} \text{ is } \begin{pmatrix} 4 \\ 2 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} -3 \\ 3 \\ 3 \end{pmatrix} = \begin{pmatrix} 4 - 3\lambda \\ 2 + 3\lambda \\ 3\lambda \end{pmatrix}$ $\vec{CM} = \begin{pmatrix} 4 - 3\lambda \\ 2 + 3\lambda \\ 3\lambda \end{pmatrix} - \begin{pmatrix} 1 \\ 4 \\ -2 \end{pmatrix} = \begin{pmatrix} 3 - 3\lambda \\ 3\lambda - 2 \\ 3\lambda + 2 \end{pmatrix}$ $\vec{AB} \Rightarrow \begin{pmatrix} 3 - 3\lambda \\ 3\lambda - 2 \\ 3\lambda + 2 \end{pmatrix} \cdot \begin{pmatrix} -3 \\ 3 \\ 3 \end{pmatrix} = 0$ Perpendicular to $\Rightarrow -9 + 9\lambda + 9\lambda - 6 + 9\lambda + 6 = 0$ $\Rightarrow 27\lambda = 9 \Rightarrow \lambda = \frac{1}{3}$ Coordinates of M are (3, 3, 1) Alternative method for last two marks Minimise	M1 (AO3.1a) A1 (AO 1.1) M1 (AO 1.1) A1 (AO 1.1) Alternative method for last two marks M1 A1 [4]	Attempt to find general point on AB to get vector CM. Can use (1, 5, 3) Allow working throughout $\begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}$ that uses e.g. $\begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}$ ft their vector from (i). Use of dot product to solve Do not accept a vector answer Alternative method for last two marks Express as a function of λ and minimise the quadratic in λ	



Question			Answer/Indicative content	Marks	Part marks and guidance	
			$\begin{vmatrix} \sqrt{r} \\ CM \end{vmatrix}^2 = \begin{vmatrix} 3-3\lambda \\ 3\lambda-2 \\ 3\lambda+2 \end{vmatrix}^2$ $= (3-3\lambda)^2 + (3\lambda-2)^2 + (3\lambda+2)^2$ $\Rightarrow \lambda = \frac{1}{3} \Rightarrow \text{Coordinates of}$ $M \text{ are } (3, 3, 1)$			



Question		Answer/Indicative content	Marks	Part marks and guidance	
	c	$CM^2 = 2^2 + 1^2 + 3^2 = 14$ $AB^2 = 3^2 + 3^2 + 3^2 = 27$ $\Rightarrow \text{Area} = \frac{1}{2} \vec{AB} \vec{CM} $ $= \frac{3}{2} \sqrt{42}$ Alternative method 1 $\text{Area} = \frac{1}{2} \vec{AB} \times \vec{BC} = \frac{1}{2} \begin{vmatrix} 3 & 0 \\ -3 & 1 \\ -3 & 5 \end{vmatrix} = \frac{1}{2} \begin{vmatrix} -12 \\ -15 \\ 3 \end{vmatrix}$ $= \frac{1}{2} \sqrt{12^2 + 15^2 + 3^2} = \frac{1}{2} \sqrt{378} = \frac{3}{2} \sqrt{42}$ Alternative method 2 $\text{Area} = \frac{1}{2} \vec{AB} \vec{BC} \sin \theta$ where $\vec{AB} \cdot \vec{BC} = \vec{AB} \vec{BC} \cos \theta$ $\Rightarrow \cos \theta = \frac{-12}{\sqrt{27} \sqrt{26}} = \frac{-4}{\sqrt{78}}$ $\Rightarrow \sin \theta = \sqrt{1 - \frac{8}{39}} = \frac{\sqrt{31}}{\sqrt{39}}$ $\Rightarrow \text{Area} = \frac{1}{2} \sqrt{27} \sqrt{26} \frac{\sqrt{31}}{\sqrt{39}} = \frac{3}{2} \sqrt{42}$	B1 (AO 1.1) B1 (AO 1.1) M1 (AO 3.1a) A1 (AO 1.1) Alternative method 1 Alternative method 1 M1 M1 A1 A1 Alternative method 2 M1 M1 A1 A1 [4]	B1 for each distance ft <i>their</i> M ft <i>their</i> AB Formula for area ft <i>their</i> M Alternative method 1 Formula for area Cross product Alternative method 2 For use of dot product, formula for area Pythagoras to find $\sin \theta$	
		Total	11		

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Question		Answer/Indicative content	Marks	Part marks and guidance		
2	a	$\left(\begin{pmatrix} 1 \\ -3 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 2 \\ -2 \end{pmatrix} \right) \cdot \begin{pmatrix} 2 \\ -5 \\ -3 \end{pmatrix} = 4$ $2 + 15 - 9 + \lambda(6 - 10 + 6) = 4$ $8 + 2\lambda = 4 \Rightarrow 2\lambda = -4 \Rightarrow \lambda = -2 \text{ so}$ $\mathbf{r} = \begin{pmatrix} 1 \\ -3 \\ 3 \end{pmatrix} + -2 \begin{pmatrix} 3 \\ 2 \\ -2 \end{pmatrix} = \begin{pmatrix} -5 \\ -7 \\ 7 \end{pmatrix}$	M1 (AO1.1) M1 (AO1.1) A1 (AO1.1) [3]	Substituting the expression for a point on the line into the equation of the plane Dotting out to form and solve equation in λ	Condone coordinates	
	b	$\frac{\begin{pmatrix} 2 \\ -5 \\ -3 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ 2 \\ -2 \end{pmatrix}}{\sqrt{4+25+9}\sqrt{9+4+4}} \text{ soi}$ $= \frac{6-10+6}{\sqrt{38}\sqrt{17}} = \frac{2}{\sqrt{646}} = 0.07868...$ $\theta = \text{awrt } 85.5...^\circ \text{ soi}$ $(\phi = 90^\circ - 85.48...^\circ =) \text{ awrt } 4.51^\circ$	M1 (AO1.1) A1 (AO1.1) A1 (AO1.1) [3]	BC. Using $\cos \theta = \frac{\mathbf{a} \cdot \mathbf{b}}{ \mathbf{a} \mathbf{b} }$ Can be implied by correct final answer	May see $\sin \phi = \frac{\mathbf{a} \cdot \mathbf{b}}{ \mathbf{a} \mathbf{b} }$ Or use of cross product or 1.49... rads or 0.0788 rads	



Question			Answer/Indicative content	Marks	Part marks and guidance		
	c		$\lambda = 1 \Rightarrow \mathbf{r} = \begin{pmatrix} 4 \\ -1 \\ 1 \end{pmatrix}$ $\mathbf{b} = \begin{pmatrix} 3 \\ 2 \\ -2 \end{pmatrix} \times \begin{pmatrix} 2 \\ -5 \\ -3 \end{pmatrix} = \begin{pmatrix} -16 \\ 5 \\ -19 \end{pmatrix}$ So equation of l_2 is $\mathbf{r} = \begin{pmatrix} 4 \\ -1 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} -16 \\ 5 \\ -19 \end{pmatrix}$ oe	B1 (AO3.1a) M1 (AO2.2a) A1 (AO1.1) [3]	Method shown or at least two terms correctly evaluated Must be r =. Allow parameter λ .		
			Total	9			



Question		Answer/Indicative content	Marks	Part marks and guidance		
3	a	$\left \begin{pmatrix} 4 \\ -5 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ 6 \\ -2 \end{pmatrix} - 15 \right $ $\left \begin{pmatrix} 3 \\ 6 \\ -2 \end{pmatrix} \right $ $\left \begin{pmatrix} 3 \\ 6 \\ -2 \end{pmatrix} \right = \sqrt{3^2 + 6^2 + 2^2} \quad \text{or}$ $\begin{pmatrix} 4 \\ -5 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ 6 \\ -2 \end{pmatrix} = 4 \times 3 + (-5) \times 6 + 1 \times (-2)$	M1(AO1.1a) M1(AO1.1) A1(AO1.1)[3]	Substitution into formula. Condone missing mod signs on top Soi	Numerator can be q. $\left \mathbf{q} \cdot \begin{pmatrix} 3 \\ 6 \\ -2 \end{pmatrix} \right $ where q is vector from C to point on // Or $\mathbf{r} = \begin{pmatrix} 4 \\ -5 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 6 \\ -2 \end{pmatrix}$ substitute into equation of // to find a value for λ.	

Examiner's Comments

The formula for shortest distance from a point to a plane is given in the formulae booklet and most candidates successfully used it, however a small number used the wrong vector for the denominator. Some candidates selected alternative valid methods. The most common of these was to substitute into the equation of the plane a general point on the normal from C; although this generally proved successful, it generally took more time.



Question		Answer/Indicative content	Marks	Part marks and guidance			
	b	$\mathbf{r} = \begin{pmatrix} 5 \\ 2 \\ 4 \end{pmatrix} - \begin{pmatrix} 4 \\ 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix}$ $\begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} = \sqrt{11}\sqrt{6}\cos\theta$ $\cos\theta = \pm \frac{6}{\sqrt{66}}$ $d = \mathbf{r} \sin\theta = \mathbf{r} \sqrt{1 - \cos^2\theta}$ $\sqrt{5}$ Alternate method 1 $\mathbf{r} = \begin{pmatrix} 5 \\ 2 \\ 4 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} - \begin{pmatrix} 4 \\ 3 \\ 1 \end{pmatrix}$ $\left(\begin{pmatrix} 5 \\ 2 \\ 4 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} - \begin{pmatrix} 4 \\ 3 \\ 1 \end{pmatrix} \right) \cdot \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} = 0$ $(\mu = -1 \text{ so}) \begin{pmatrix} 4 \\ 4 \\ 3 \end{pmatrix}$ $d = \sqrt{(4-4)^2 + (3-4)^2 + (1-3)^2}$ $\sqrt{5}$ Alternate method 2 $\mathbf{r} = \begin{pmatrix} 5 \\ 2 \\ 4 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} - \begin{pmatrix} 4 \\ 3 \\ 1 \end{pmatrix}$ $ \mathbf{r} = \sqrt{6\mu^2 + 12\mu + 11}$ $\mu = -1$ $d = \sqrt{5}$	M1*M1 A1 dep*M1 A1 M1(A O3.1a) * M1(AO1. 1a) A1(AO1. 1) dep*M 1(AO1.1) A1(AO1. 1) M1 *M1 A1 dep*M1 A1[5]	Finding the vector $\mathbf{r}$ from a point on one line to a point on the other Using dot product to find an expression for $\cos\theta$ can be implied by $\cos\theta = 0.739$, or $\theta = 0.740$ or 2.402 or 42.4° or 137.6° Finding the vector from a particular point on one line to a general point on the other Dotting the vector found with one of the direction vectors and setting to 0 Solving to find (a parameter value and hence) position vector or coordinates of	NB Ignore attempts to use the formula for the distance between 2 skew lines. or ($\lambda = -\frac{1}{2}$ and) $\lambda = -\frac{1}{2}$ As above Allow if full method seen or if 2 terms correct		

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Question			Answer/Indicative content	Marks	Part marks and guidance		
			Alternate method 3 $\mathbf{r} = \begin{pmatrix} 5 \\ 2 \\ 4 \end{pmatrix} - \begin{pmatrix} 4 \\ 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix}$ $\left \begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix} \times \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} \right = \left \begin{pmatrix} 5 \\ 2 \\ -1 \end{pmatrix} \right $ $\dots = \sqrt{5^2 + 2^2 + 1^2} = \sqrt{30}$ $d = \frac{\sqrt{30}}{\left \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} \right } = \sqrt{5}$		equivalent point on other line Distance formula for their 2 points Finding the vector from a particular point on one line to a general point on the other Attempts to find parameter (Calculus or completes square) Finding the vector r from a point on one line to a point on the other Calculates cross product of 'their r' and direction vector. <u>Examiner's Comments</u> To find the distance between parallel lines is an explicitly stated part of the specification, yet many candidates appeared unfamiliar with the topic. A significant number started by attempting to use the formula for distance		

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Question			Answer/Indicative content	Marks	Part marks and guidance	
					between skew lines. If they were successful in performing the cross-product they usually realised that the lines were parallel, but quite a few candidates then did not know how to proceed. Some simply found a vector from one line to the other and reasoned that its magnitude must be the required distance, even though they had made no attempt to make sure that it was perpendicular to the lines. Others formed planes and found the distance between them, but these were often the wrong planes. There were nonetheless a number of different successful ways found to reach the solution. Where candidates did identify a correct approach to the problem they generally came up with a good response, barring arithmetic errors. The main problem that these candidates had was ensuring that their answer was an exact value (as was requested). Some candidates appeared unable to convert cosine to sine and presented a rounded answer. Others presented their answer using the form $\sin(\cos^{-1}p)$, which was not acceptable for a final answer.	
			Total	8		



Question		Answer/Indicative content	Marks	Part marks and guidance		
4	a	$\begin{pmatrix} 13 \\ 3 \\ -14 \end{pmatrix} = \begin{pmatrix} 1 \\ 5 \\ 3 \end{pmatrix} + \mu \begin{pmatrix} 7 \\ 9 \\ -2 \end{pmatrix} \Rightarrow 13 = 1 + 7\mu \Rightarrow \mu = 2$ (so R is on l_1) $\text{eg } 7 - \mu = 13 \Rightarrow \mu = -6 \Rightarrow$ $\mathbf{r} = \begin{pmatrix} 7 \\ 9 \\ -2 \end{pmatrix} - 6 \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 13 \\ 3 \\ -14 \end{pmatrix}$ (so R is also on l_2) Alternate method $\begin{pmatrix} 1 \\ 5 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 7 \\ 9 \\ -2 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix} + \mu \begin{pmatrix} 7 \\ 9 \\ -2 \end{pmatrix} \Rightarrow 46 + 10\mu = -14 \Rightarrow \mu = -6$ $\mu = -6 \Rightarrow$ $\mathbf{r} = \begin{pmatrix} 7 \\ 9 \\ -2 \end{pmatrix} - 6 \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 13 \\ 3 \\ -14 \end{pmatrix}$ so R is (13, 3, -14)	B1 (AO1.1) B1 (AO1.1) M1 A1 [2]	AG. Intermediate working must be seen AG. Or $9 + \mu = 3$ or $-2 + 2\mu = -14$ but must be checked in other two equations. AG. Substituting in expression of the point into the equation of the plane to find a value for μ AG.	Answer in vector form is acceptable.	
	b	Since lines intersect $\begin{pmatrix} 0 \\ 1 \\ a \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ -3 \end{pmatrix} = \begin{pmatrix} 7 \\ 9 \\ -2 \end{pmatrix} + \mu \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix}$ for some λ and μ so $1 + 2\lambda = 7 - \mu$ $\lambda = 9 + \mu$ $(a - 3\lambda - 2 + 2\mu)$ $\Rightarrow \lambda = 5, \mu = -4$ $\text{so } a + 5 \times (-3) = -2 + (-4) \times 2 \Rightarrow a = 5$ At for some λ	M1 (AO3.1a) A1 (AO1.1) A1ft (AO1.1) (AO2.1) *M1	Equating the lines and deriving 2 useful equations. Ignore attempts at z coefficient equation	Can be BC	



Question			Answer/Indicative content	Marks	Part marks and guidance		
			$Q, \begin{pmatrix} 1 \\ 0 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ -3 \end{pmatrix} = \begin{pmatrix} 1 \\ 5 \\ 3 \end{pmatrix} \Rightarrow -14$ so $1 + 5 \times 3 + \lambda(2 + 5 + (-3) \times 3) = -14$ $-2\lambda = -30 \Rightarrow \lambda = 15 \Rightarrow Q(31, 15, -40)$ $\sqrt{(31-13)^2 + (15-3)^2 + (-40-14)^2}$ $= \sqrt{18^2 + 12^2 + 26^2} = \sqrt{1144}$	A1 (AO1.1) dep*M1 (AO2.1) A1 (AO3.2a) [7]	Substituting general r from l_1 into the l_2 equation and dotting out to form an equation in λ Method fully shown or at least 2 of 3 squared terms correct 2√286 Examiner's Comments The command word 'verify', in (a), should act as a trigger to attempt to 'show it works by substitution'. Many candidates, instead, used the correct method to find an unknown intersection point. While this gained full credit, centres should make sure that candidates take advantage of the fact that verifying is usually easier than deriving.	Accept algebraic expressions in a until final A mark $\lambda = 7.5 + 1.5a$ Q(16 + 3a, 7.5 + 1.5a, -22.5 - 3.5a) Depends on correct method shown to find Q $\sqrt{(3+3a)^2 + (4.5+1.5a)^2 + (-8.5-3.5a)^2}$	
			Total	9			



Question			Answer/Indicative content	Marks	Part marks and guidance	
5			$\mathbf{n} = \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix} \times \begin{pmatrix} 1 \\ -1 \\ 4 \end{pmatrix} = \begin{pmatrix} 2 \\ -10 \\ -3 \end{pmatrix}$ $ \mathbf{n} = \sqrt{113}$ $\mathbf{b} - \mathbf{a} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} - \begin{pmatrix} 2 \\ 2 \\ -3 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \\ 2 \end{pmatrix}$ $\Rightarrow d = \frac{\left \begin{pmatrix} -1 \\ 0 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ -10 \\ -3 \end{pmatrix} \right }{ \mathbf{n} } = \frac{8}{\sqrt{113}} = 0.753$ to 3sf	M1 (AO3.1a) A1 (AO1.1) A1 (AO1.1) M1 (AO1.1) A1 (AO1.1) [5]	Cross product Modulus Using correct formula for d Accept exact or correct to 3sf (0.75257669...) FT an exact answer <u>Examiner's Comments</u> For this question not only was the formula for the distance required but also the cross product of the directions of the lines and the dot product of two position vectors, one on each line. A significant number of candidates made mistakes in their calculations even when there was evidence that the correct method was known.	
			Total	5		

$\Rightarrow 16\lambda^2 + 4\lambda^2 + \lambda^2 = 336 \Rightarrow$ $21\lambda^2 = 336$ $\Rightarrow \lambda^2 = 16$ M1 $\Rightarrow \lambda = \pm 4$ A1 (25, 14, 5) is the same side	to double that distance to get to the reflection point; about half of these candidates managed to obtain the correct answer. The question asked for the
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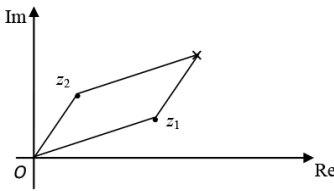
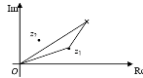


Question			Answer/Indicative content	Marks	Part marks and guidance
					Exemplar 1 Coordinates of $P: a$ Normal to plane = $\begin{pmatrix} 4 \\ 2 \\ 1 \end{pmatrix}$ $\therefore P = \begin{pmatrix} 9 \\ 6 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ 2 \\ 1 \end{pmatrix}$ $4x + 2y + z = 7$ $4(9 + 4\lambda) + 2(6 + 2\lambda) + 1(1 + \lambda) = 7$ $36 + 16\lambda + 12 + 4\lambda + 1 + \lambda = 7$ $49 + 21\lambda = 7$ $21\lambda = -42$ $\lambda = -2$ $B = \begin{pmatrix} 9 \\ 6 \\ 1 \end{pmatrix} + 2\lambda \begin{pmatrix} 4 \\ 2 \\ 1 \end{pmatrix}$ $= \begin{pmatrix} 9 \\ 6 \\ 1 \end{pmatrix} - 4 \begin{pmatrix} 4 \\ 2 \\ 1 \end{pmatrix}$ $= \begin{pmatrix} -7 \\ -2 \\ -3 \end{pmatrix} \rightarrow B = (-7, -2, -3)$ midpoint = $(1, 2, -1)$ ✓ This candidate has not found the need to calculate distances, but has used the property that the intersection of the normal line from A with the plane is the midpoint of the line AB. Full marks are earned for this.
			Total	6	

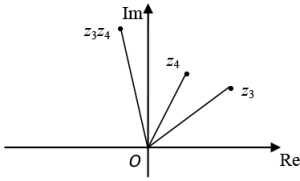
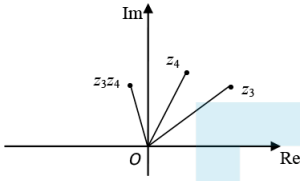


Question			Answer/Indicative content	Marks	Part marks and guidance	
7		a	$AB = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}, AC = \begin{pmatrix} 3 \\ 1 \\ 9 \end{pmatrix}$ $\Rightarrow AB \times AC = \begin{pmatrix} -2 \\ -3 \\ 1 \end{pmatrix}$	B1 (AO 1.1) M1 (AO 1.1a) A1 (AO 1.1) [3]	Either correct Cross product BC	
		b	$-2x - 3y + z = d$ $\Rightarrow$ e.g. $-2 \times 0 - 3 \times 1 - 1 \times 4 = -7$ $\Rightarrow 2x + 3y - z = 7$ oe	M1 (AO 3.1a) A1 (AO 1.1) [2]	Use of <i>their</i> vector product and substitution of one point	
			Total	5		



Question			Answer/Indicative content	Marks	Part marks and guidance	
8		a		B1 (AO 2.2a) B1 (AO 2.2a) [2]	2 lines drawn as shown to complete a parallelogram Or 2 lines drawn to form a triangle which is either the upper or lower half of the parallelogram (split by the leading diagonal). eg 	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		b	(i) 	B1 (AO 1.1)	z_3 and z_4 are approximately correctly positioned and labelled.	If no labels shown then B1B1 can only follow if there is no ambiguity between points (eg magnitudes shown).
			(ii) 	[2] B1 (AO 1.1)	Approximate correct length (eg z_4 length increased by 50%) and angle (about a quarter of the way round the 2nd quadrant).	$r = 1.8, \theta = \frac{5}{8}\pi$
				B1 (AO 2.2a)	z_3 and z_4 are approximately correctly positioned and labelled.	If no labels shown then B1B1 can only follow if there is no ambiguity between points (eg magnitudes shown).
				[2]	Approximate correct length (eg z_3 length halved) and either the same angle as part (b)(i) or about a quarter of the way round the 2nd quadrant.	$r = 0.35, \theta = \frac{5}{8}\pi$
			Total	6		



Question			Answer/Indicative content	Marks	Part marks and guidance		
9		a	$\mathbf{n}_1 \cdot \mathbf{n}_2 = 9 = \mathbf{n}_1 \mathbf{n}_2 \cos \theta = \sqrt{14}\sqrt{6} \cos \theta$ $\Rightarrow \cos \theta = \frac{9}{\sqrt{84}}$ $\Rightarrow \theta \approx 10.9^\circ \text{ or } 0.190 \text{ rads}$	M1 (AO1.1a) A1 (AO1.1) A1 (AO1.1) [3]		soi	$\mathbf{n}_1 = \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix}$ and $\mathbf{n}_2 = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}$ d



Question			Answer/Indicative content	Marks	Part marks and guidance		
		b	Direction of line is $\begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix}$ $\mathbf{n}_1 \cdot \begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix} = 0$ and $\mathbf{n}_2 \cdot \begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix} = 0$ Point (0, 1, 2) is on L: $\Pi_1: 3 \times 0 + 2 \times 1 + 2 = 0 + 2 + 2 = 4 \checkmark$ $\Pi_1: 2 \times 0 + 1 + 2 = 3 \checkmark$ So a point on L lies in both planes. But L is parallel to both planes so L lies in both planes. OR	B1 (AO2.2a) M1 (AO3.1a) A1 (AO1.1) M1 (AO3.1a) A1 (AO3.2a) B1 M1 A1	direction Scalar product with both normals (shown, not just stated) finding zero twice A point on the line into both equations showing it is in both planes convincingly and conclusion Substituting	OR M1 for attempt to find vector product of the normals A1 for showing it to be a multiple of the direction vector	



Question			Answer/Indicative content	Marks	Part marks and guidance		
			$r = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} \text{ or } \begin{pmatrix} \lambda \\ 1-\lambda \\ 2-\lambda \end{pmatrix}$ $\Pi_1: \mathbf{r} \cdot \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} \lambda \\ 1-\lambda \\ 2-\lambda \end{pmatrix} \cdot \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix}$ $= 3\lambda + 2 - 2\lambda + 2 - \lambda = 4$ $\Pi_2: \mathbf{r} \cdot \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} \lambda \\ 1-\lambda \\ 2-\lambda \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}$ $= 2\lambda + 1 - \lambda + 2 - \lambda = 3$ So L lies in both planes. Or $y = 1 - x, z = 2 - x$ $\Pi_1: 3x + 2y + z = 3x + 2(1 - x) + 2 - x$ $= 3x + 2 - 2x + 2 - x = 4$ $\Pi_2: 2x + y + z = 2x + 1 - x + 2 - x$ $= 3$ So L lies in both planes.	M1 A1 B1 M1 A1 M1 A1 [5]	into equation of Π_1 or Π_2 Substituting into other plane equation Correct values and conclusion Expressing two of x, y and z correctly in terms of the other Substituting into equation of Π_1 Substituting into equation of Π_2 Correct values and conclusion.		
			Total	8			



Question			Answer/Indicative content	Marks	Part marks and guidance	
10		a	$\begin{pmatrix} 1 \\ 1 \\ 3 \end{pmatrix} \times \begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix} = \dots$ $\dots \begin{pmatrix} -5 \\ -4 \\ 3 \end{pmatrix}$	M1 (AO 1.1a) A1 (AO 1.1) [2]	Attempt to find cross product (can be implied by one correct component (or all 3 with wrong sign)) Or any non-zero multiple.	
		b	$\mathbf{r} \cdot \begin{pmatrix} -5 \\ -4 \\ 3 \end{pmatrix} = \begin{pmatrix} 2 \\ -3 \\ 5 \end{pmatrix} \cdot \begin{pmatrix} -5 \\ -4 \\ 3 \end{pmatrix} = 2 \times -5 + -3 \times -4 + 5 \times 3$ $\mathbf{r} \cdot \begin{pmatrix} -5 \\ -4 \\ 3 \end{pmatrix} = 17$	M1 (AO 1.1a) A1 (AO 1.1) [2]	Use their n from part (a) in $\mathbf{r} \cdot \mathbf{n} = \mathbf{a} \cdot \mathbf{n}$ and attempt to find dot product. "Dot product" must be a scalar. Or any non-zero multiple of entire equation	



Question			Answer/Indicative content	Marks	Part marks and guidance		
		c	$\left \begin{pmatrix} -5 \\ -4 \\ 3 \end{pmatrix} \right = \sqrt{25+16+9} = \sqrt{50}$ $d = \frac{17}{\sqrt{50}} \text{cao}$	M1 (AO 1.1) A1 (AO 2.2a) [2]	Correct method for finding modulus of their normal vector Must be positive		
			Total	6			



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Question			Answer/Indicative content	Marks	Part marks and guidance		
11		a	$\frac{x(+0)}{-2} = \frac{y-5}{2} = \frac{z+(-6)}{-7}$ seen or implied by a or d correct $\mathbf{r} = \begin{pmatrix} 0 \\ 5 \\ -6 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 2 \\ -7 \end{pmatrix}$ or	M1 (AO 1.1a) A1 (AO 1.1) [2]	Expressing in form from which a and d from $\mathbf{r} = \mathbf{a} + \lambda \mathbf{d}$ can be read off. Must be $\mathbf{r} =$. Must have sensible parameter name (not eg x, y, z or r).	Or writing $\frac{-x}{2} = \lambda$ etc and attempting to solve for x, y and z	
		b	Any two of $-2\lambda = 2 + \mu$, $5 + 2\lambda = 7 - 2\mu$ and $-6 - 7\lambda = -1 + 4\mu$ attempt to solve $\lambda = -3$ or $\mu = 4$ $(6, -1, 15)$	M1 (AO 1.1a) A1 (AO 1.1) A1 (AO 1.1) [3]			



Question			Answer/Indicative content	Marks	Part marks and guidance	
		c	$\begin{pmatrix} -2 \\ 2 \\ -7 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -2 \\ 4 \end{pmatrix} (= -34)$ $\cos^{-1} \frac{\pm 34}{\sqrt{57}\sqrt{21}}$ awrt 10.7° or 0.186 rads www	M1* (AO 1.1a) dep*M1 (AO 1.1) A1 (AO 1.1) [3]	Correct method for finding (modulus of) dot product of $\begin{pmatrix} 1 \\ -2 \\ 4 \end{pmatrix}$ and their direction vector Correct process for finding moduli and correct use (ie inverse cos of $\pm$ their dot product divided by the product of their moduli). Acute angle only	
Total				8		

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Question			Answer/Indicative content	Marks	Part marks and guidance	
12		a	Find normal by vector product $\mathbf{n} = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \times \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} = \begin{pmatrix} 7 \\ -1 \\ -5 \end{pmatrix}$ Taking $\mathbf{b} = (7, 1, 6)$ and $\mathbf{a} = (2, 6, -2)$ $\mathbf{b} - \mathbf{a} = (5, -5, 8)$ Using distance formula: $(\mathbf{b} - \mathbf{a}) \cdot \mathbf{n} = \begin{pmatrix} 5 \\ -5 \\ 8 \end{pmatrix} \cdot \begin{pmatrix} 7 \\ -1 \\ -5 \end{pmatrix} = 0$ So distance = 0 so they intersect. Alternative: Any point on l_1 is $(7 + 2\lambda, 1 - \lambda, 6 + 3\lambda)$ Any point on l_2 is $(2 + \mu, 6 + 2\mu, -2 + \mu)$ Equate any two: $\Rightarrow \lambda = -3, \mu = -1$ and the point is $(1, 4, -3)$	M1 (AO1.1a) A1 (AO1.1) M1 (AO2.1) A1 (AO1.1) M1 A1 M1 A1 [4]		



Question			Answer/Indicative content	Marks	Part marks and guidance	
		b	Use of normal vector $7x - y - 5z = c$ Through a or b $\Rightarrow c = 18$ $\Rightarrow 7x - y - 5z = 18$	B1 (AO1.1) M1 (AO1.1) A1 (AO1.1) [3]		
			Total	7		



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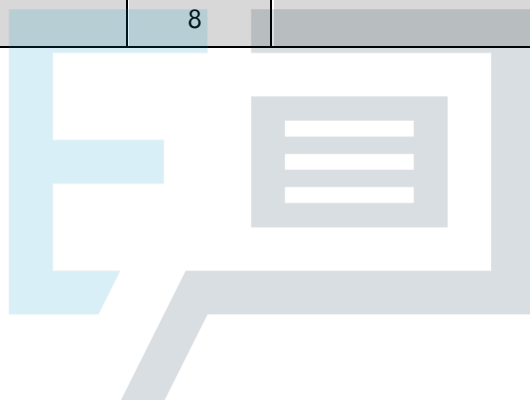
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Question			Answer/Indicative content	Marks	Part marks and guidance	
13		a	$\begin{pmatrix} 2 \\ 2 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} = 10$ $\cos(90 - \theta) = \frac{10}{\sqrt{17}\sqrt{14}} \quad (= \sin \theta)$ $(\theta = 90 - 49.6) = \text{awrt } 40.4$	M1* (AO 1.1a) M1dep* (AO 1.1) A1 (AO 1.1) [3]	Correct process for finding the dot product of normal to the plane and direction vector of line. Correct process for using dot product to find the cosine of the complement or sin of angle Or awrt 0.705	
		b	$\begin{pmatrix} 1 \\ 0 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix}$ $\lambda = 2$ $(5, 4, 11)$	M1 (AO 1.1a) A1 (AO 1.1) A1 (AO 1.1) [3]	Correct substitution line form for r into r.n = d Correct substitution line form for r into r.n = d Condone position vector	or $3(1 + 2\lambda) - 2\lambda + 2(5 + 3\lambda) = 33$



Question			Answer/Indicative content	Marks	Part marks and guidance		
		c	$\frac{ 3 \times 4 + (-1) \times 5 + 2 \times (-5) - 33 }{\sqrt{3^2 + (-1)^2 + 2^2}}$ $\frac{18\sqrt{14}}{7}$	M1 (AO 1.1a) A1 (AO 1.1) [2]	Substitution of S and the normal to the plane into the formula. $c \frac{36}{\sqrt{14}}$ r		
			Total	8			



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Question			Answer/Indicative content	Marks	Part marks and guidance		
14			Any two from $3 + \lambda = -1 - 3\mu$ $\lambda = 8 + \mu$ $2 + 3\lambda = 2 - 5\mu$ solve simultaneously to obtain a value of λ or μ $\lambda = 5$ or $\mu = -3$ (8, 5, 17) isw	B1 M1 A1 A1 [4]	may be in vector form allow vector form		
			Total	4	Examiner's Comments Most candidates scored full marks on this question. Very few were unable to make some progress, and those that went wrong usually did so through careless slips.		

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Question			Answer/Indicative content	Marks	Part marks and guidance	
15		i	$\mathbf{r} \cdot \begin{pmatrix} 2 \\ -3 \\ 5 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ -4 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ -3 \\ 5 \end{pmatrix} = -24$ $\text{distance} = \frac{7 - (-24)}{\sqrt{2^2 + 3^2 + 5^2}}$ $= \frac{31}{\sqrt{38}}$	M1 M1 A1 [3]	// plane through A Oe such as 5.03 Examiner's Comments Many candidates remain unaware of the existence of the formula for perpendicular distance from a point to a plane which is given in the List of Formulae(MF1). Those who used the standard formula were mostly successful although some got the sign of "d" wrong. Using other (lengthier) methods such as finding the parameter for the base of the perpendicular worked well for some candidates, but others made numerical errors or didn't have a complete method to hand.	ALT. $2(1 + 2\lambda) - 3(2 - 3\lambda) + 5(-4 + 5\lambda) = 7.$ $\lambda = \frac{31}{38}$ $\text{distance} = \frac{d^2 y}{dx^2} + 6 \frac{dy}{dx} + 5y - 25 \sin x.$

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Question			Answer/Indicative content	Marks	Part marks and guidance		
		ii	$\sin \theta = \frac{\left \begin{pmatrix} 2 \\ -3 \\ 5 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} \right }{\sqrt{2^2+3^2+5^2}\sqrt{1^2+1^2+2^2}}$ $\theta = 1.46 \text{ (or } 83.4^\circ)$	M1 M1 A1 [3]	For RHS Suitable method for finding required angle Examiner's Comments The standard method for finding this angle was familiar to many candidates. Even those who were unsuccessful could usually gain the first mark by finding the angle between line and normal to plane, even if they did then take this to be the solution.	0.1150	

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Question			Answer/Indicative content	Marks	Part marks and guidance		
		iii	$\mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ -4 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$ $2(1 + \lambda) - 3(2 - \lambda) + 5(-4 + 2\lambda) = 7$ $\lambda = \frac{31}{15}$ Intersect at $\square\square\square$ $\left(\frac{46}{15}, -\frac{1}{15}, \frac{2}{15}\right)$	B1 M1 A1 A1 [4]	Substitute in plane equation Or position vector. Accept (3.07, – 0.0667, 0.133)		
Total				10			



Question			Answer/Indicative content	Marks	Part marks and guidance		
16		i	$\vec{AB} = \begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ -5 \\ 2 \end{pmatrix}$ $\vec{AC} = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} - \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 0 \\ -2 \\ -2 \end{pmatrix}$ $\mathbf{n} = \begin{pmatrix} 1 \\ -5 \\ 2 \end{pmatrix} \times \begin{pmatrix} 0 \\ -2 \\ -2 \end{pmatrix} = \begin{pmatrix} -11 \\ -7 \\ 3 \end{pmatrix} = -\begin{pmatrix} 11 \\ 7 \\ -3 \end{pmatrix}$ $11x + 7y + 12z = 11(1) + 7(2) + 12(-1)$ $11x + 7y + 12z = 13$	M1* *M1dep A1 A1 [4]	Any two vectors in plane Depends on using attempted vectors in plane Condone 1 incorrect element if no working. Any multiple – linked to second M1 only Condone omission of final minus sign in this argument Must show substitution or dot product www. Shown ag. Must have some reasoning e.g. AB and AC referenced or described as a vector in the plane, normal referenced, $r = a + sb + tc$	Thir $\begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} - \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 0 \\ -1 \\ 0 \end{pmatrix}$ d is ALT $r = a + sb + tc$ Then eliminates one parameter to form 2 equations Then eliminates t to get plane (A2, with A1 awarded for each side of equation SC4 or verifying that all three points lie on the given plane and checking for non-collinearity	



Question			Answer/Indicative content	Marks	Part marks and guidance		

Examiner's Comments
Almost all candidates appeared familiar with one of the methods for deriving the equation of a plane through 3 (non-collinear) points. However, a number of candidates lost marks through failing to give sufficient justification. It is important at this level, when asked to "show that" (rather than simply to "find"), to communicate your method mathematically. Candidates were expected to explicitly show the required vector subtractions for obtaining the direction of two vectors in the plane, and to exhibit at least something of the geometrical context, such as by reference to the "normal" vector.



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$\begin{pmatrix} 11 \\ 7 \\ 12 \end{pmatrix} \times \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -5 \\ 25 \\ -10 \end{pmatrix} = -5 \begin{pmatrix} 1 \\ -5 \\ 2 \end{pmatrix}$ $x = 0 \Rightarrow y = 7, z = -3$ $\mathbf{r} = \begin{pmatrix} 0 \\ 7 \\ -3 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -5 \\ 2 \end{pmatrix}$	M1 A1 B1 A1 [4]	Attempts cross product of correct vectors Any multiple Find a point on line Oe vector equation ALT 1: Find a point on line M1 Find a second point and use to find direction of line M1, A1 Write equation A1 Examiner's Comments Many candidates proficiently found the line of intersection; the usual method being to calculate the cross product and find a particular point. Some did it by eliminating a variable from the pair of plane equations and then writing x, y, z in terms of a parameter, but this approach was more prone to errors of calculation or errors when converting their parametric form into vector equation form.	or $\begin{pmatrix} 7 \\ 5 \\ 0 \\ 1 \\ 5 \end{pmatrix}, \text{ or } \begin{pmatrix} 3 \\ 2 \\ 1 \\ 2 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ -1 \\ 2 \\ 1 \end{pmatrix}$ A2: Reduce 2 equations to single equation in 2 variables. M1 Write these 2 variables using a parameter. M1 Find third variable parametrically. A1 Write equation. A1



Question			Answer/Indicative content	Marks	Part marks and guidance		
		iii	$\cos \theta = \frac{\left \begin{pmatrix} 11 \\ 7 \\ 12 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix} \right }{\sqrt{11^2 + 7^2 + 12^2} \sqrt{3^2 + 1^2 + 1^2}}$ $\theta = 0.485 \text{ (or } 27.8^\circ)$	M1 A1 [2]		0/2 for $90 - \theta$	Examiner's Comments The angle required was usually correctly found. The only notable error was to give the complementary angle, either by misuse of sine in the scalar product or by confusing the method required here with the one for finding the angle between a line and a plane. A few candidates when finding angles in this paper insisted on using the lengthier method of using the vector product rather than the scalar product and, although this is acceptable, centres may wish to stress the time-saving benefits of the latter.
			Total	10			



Question			Answer/Indicative content	Marks	Part marks and guidance		
17		a	$\mathbf{n} = \begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix} \times \begin{pmatrix} 2 \\ -1 \\ 4 \end{pmatrix} = \begin{pmatrix} 5 \\ -10 \\ -5 \end{pmatrix}$ $\mathbf{b} - \mathbf{a} = \begin{pmatrix} 4 \\ -2 \\ 7 \end{pmatrix} - \begin{pmatrix} 3 \\ 5 \\ -2 \end{pmatrix} = \begin{pmatrix} 1 \\ -7 \\ 9 \end{pmatrix}$ Shortest distance is given by $D = \frac{ (\mathbf{b} - \mathbf{a}) \cdot \mathbf{n} }{ \mathbf{n} } = \frac{\left \begin{pmatrix} 1 \\ -7 \\ 9 \end{pmatrix} \cdot \begin{pmatrix} 5 \\ -10 \\ -5 \end{pmatrix} \right }{\sqrt{150}}$ $= \sqrt{6}$	M1(AO 1.1a) A1(AO 1.1) B1(AO 1.1) M1(AO 1.2) A1(AO 1.1) [5]	Finding a mutual perpendicular Any vector between the lines Using formula from the formula booklet	Accept any multiple Or any other valid method	
		b	$x - 2y - z = -5$	M1(AO 3.1a) A1(AO 1.1) [2]	Using their mutual perpendicular from part (i) or		
			Total	7			



Question			Answer/Indicative content	Marks	Part marks and guidance		
18		a	$-5 - 6 - 2 = \sqrt{1+4+1}\sqrt{25+9+4} \cos \theta$ oe $\theta = \cos^{-1} \frac{-13}{\sqrt{6 \times 38}}$ oe $149.4^\circ - 90^\circ$ 59.4°	*M1(AO1.1a) dep*M1(AO1.1) M1(AO2.2a) A1(AO1.1) [4]	Allow one slip e.g. sign error Using angle with normal to find angle with plane 1.04 in radians		
		b	$1 \times 3 + (-2) \times (-1) + (-1) \times 5 = 0$ The line is in or parallel to the plane $1 \times 1 - 2 \times 4 - 1 \times (2) \neq 30$ Point on line not in plane, so line is parallel to plane	M1(AO3.1a) E1(AO2.2a) M1(AO3.1a) E1(AO2.2a) [4]	Alternative method $1 + 3\mu - 2(4 - \mu) - (2 + 5\mu) = -9$ M1A1 -9 is not 30 E1 Conclusion E1		
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Question			Answer/Indicative content	Marks	Part marks and guidance	
19		i	$3 + 5t = 1 + 2s$ $2 - 3t = 4 - s$ $-5 + t = 5 + 3s$	M1	attempt to solve any two of these simultaneously to obtain a value of s or t	
		i	$t + -2$ and $s = -4$	A1		
		i	substitution of their s and t in other equation to obtain eg $-7 = -7$ oe (1 st or 3 rd equation) or eg $8 = 8$ oe (2 nd equation)	B1	may be embedded, eg $-5 + -2 = 5 + 3 \times -4$	B0 if any subsequent arithmetic errors seen
		i	lines meet at $(-7, 8, -7)$	A1	allow in vector form	eg $-5 + -2 = 5 + 3 \times -4$ so $7 = -7$
		ii	$\begin{pmatrix} -4 \\ 2 \\ -6 \end{pmatrix} = -2 \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$ oe seen	B1	dot not allow eg $\begin{pmatrix} -4 \\ 2 \\ -6 \end{pmatrix} \div \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} = -2$	or $\begin{pmatrix} 7 \\ 1 \\ 14 \end{pmatrix} - \begin{pmatrix} 1 \\ 4 \\ 5 \end{pmatrix} = \begin{pmatrix} 6 \\ -3 \\ 9 \end{pmatrix}$
		ii	common point identified and justified eg by substitution of correct value of s or u eg $s = 3$ or eg $u = \frac{3}{2}$	B1		$= 3 \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$
		ii	Alternatively		or eg $u = \frac{3}{2}s - \frac{1}{2}$	or
		ii	substitution of eg $s = 3 - 2u$	B1	and completion to $r = \begin{pmatrix} 1 \\ 4 \\ 5 \end{pmatrix} + u \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$	show one pair of equations consistent



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	and completion to $\mathbf{r} = \begin{pmatrix} 7 \\ 1 \\ 14 \end{pmatrix} + u \begin{pmatrix} -4 \\ 2 \\ -6 \end{pmatrix}$	B1		show another pair consistant Examiner's Comments Part (i) was done very well indeed. Most candidates found s and t successfully and went on to find the point of intersection and confirm that all three equations were consistent. A few candidates lost an easy mark by omitting one of the last two requirements. In part (ii) many candidates successfully demonstrated the equivalence of the direction vectors, but rather fewer appreciated the need to identify a common point. Various other approaches were seen but in many cases candidates left rather too much to the imagination of the examiner to earn full marks.
			Total	6		



Question			Answer/Indicative content	Marks	Part marks and guidance	
20			$\begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} \times \begin{pmatrix} 3 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ -4 \\ -6 \end{pmatrix} = -2 \begin{pmatrix} -1 \\ 2 \\ 3 \end{pmatrix}$	M1	at least 2 correct values for the cross product or method shown	
			$\begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} - \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 3 \\ 0 \\ -2 \end{pmatrix}$	A1	Any multiple	
			Shortest distance $= \frac{\left \begin{pmatrix} 3 \\ 0 \\ -2 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ 2 \\ 3 \end{pmatrix} \right }{\sqrt{1^2 + 2^2 + 3^2}}$	M1		
			$= \frac{9}{\sqrt{14}} \text{ or } 2.41$	A1		
						Examiner's Comments The majority of candidates had learnt the standard method given in the mark scheme for finding the shortest distance between lines and, apart from minor slips, they were able to handle this question with ease. Another fairly common, successful approach was to convert the problem into one of finding the shortest distance between two parallel planes each containing one of the lines. A valid alternate approach which was sometimes seen involved using 3 parameters – one for each line plus one for the vector joining them perpendicularly. Most candidates attempting this approach were unable to complete it accurately. Some candidates attempted to use the standard method, but without secure knowledge of which vectors went where within the formula.



Question			Answer/Indicative content	Marks	Part marks and guidance
			Total	4	



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Question			Answer/Indicative content	Marks	Part marks and guidance	
21		i	$3 \sin \alpha \times 2 \cos \alpha \times 2 \cos \alpha \times 4 \sin \alpha + -1 \times 3$	M1		allow one sign error or one coefficient error for M1
		i	$6 \sin \alpha \cos \alpha + 8 \sin \alpha \cos \alpha - 3 = 0$ so i	A1		
		i	substitution of $\sin \alpha \cos \alpha = \frac{1}{2} \sin 2\alpha$ oe	M1	NB $7 \sin 2\alpha = 3$	or squaring both sides and correct substitution from Pythagoras
		i	$\alpha = \text{awrt } 12.7^\circ$	A1	awrt 0.221	
		i	$\alpha = \text{awrt } 77.3^\circ$	A1	awrt 1.35	if A0A0, SC1 for 13° and 77° or 0.22 and 1.4
Examiner's Comments Nearly all candidates understood how to use the scalar product to solve this problem, and many went on to score full marks. A few lost an accuracy mark because they were unable to find the larger angle correctly, and some failed to spot that a double angle formula was required.						
		ii	their $\alpha = 12.7^\circ$ substituted in $\vec{OA}$ and $\vec{OB}$; or in $\frac{ \vec{OA} }{ \vec{OB} }$ 	M1		NB $\begin{pmatrix} 0.6589.. \\ 1.9511.. \\ -1 \end{pmatrix}$ and $\begin{pmatrix} 1.9511.. \\ 0.8785.. \\ 3 \end{pmatrix}$
		ii	$\sqrt{(3 \sin \alpha)^2 + (2 \cos \alpha)^2 + (-1)^2}$ or $\sqrt{(2 \cos \alpha)^2 + (4 \sin \alpha)^2 + 3^2}$	M1*	allow omission of brackets, one slip in arithmetic and one sign error;	
		ii	$\frac{1}{2} \sqrt{9 \sin^2 \alpha + 4 \cos^2 \alpha + 1} \sqrt{4 \cos^2 \alpha + 16 \sin^2 \alpha + 9}$	M1*dep	may be implied by numerical value for lengths; allow one sign or coefficient error α may be unspecified or any acute angle for these method marks	NB $\sqrt{5.241..} = 2.289...$ and $\sqrt{13.579..} = 3.685...$



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	awrt 4.22	A1		NB hypotenuse is 4.34 and other angles in triangle are 58.2° and 31.8° Examiner's Comments This proved more demanding. Many candidates found the correct lengths and then automatically used $\frac{1}{2}ab\sin C$. Some lost accuracy through working with rounded numbers, or through bracket or sign errors in substitution, but a significant minority went on to score full marks. Some candidates showed very little working and achieved a wrong answer, thus denying the possibility of earning any method marks.
			Total	9		

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Question			Answer/Indicative content	Marks	Part marks and guidance	
22		i	$k(4\mathbf{i} + 8\mathbf{j} - 8\mathbf{k})$ oe seen	B1	allow eg (4, 8, -8)	NB $\overrightarrow{AB} = 4\mathbf{i} + 8\mathbf{j} - 8\mathbf{k}$ $\overrightarrow{AP} = 3\mathbf{i} + 6\mathbf{j} - 6\mathbf{k}$ $\overrightarrow{BP} = \mathbf{i} + 2\mathbf{j} - 2\mathbf{k}$
		i	$[\overrightarrow{CP} =] -2\mathbf{i} + 2\mathbf{j} + \mathbf{k}$ or $[\overrightarrow{PC} =] 2\mathbf{i} - 2\mathbf{j} - \mathbf{k}$	B1	allow eg (-2, 2, 1)	NB P is (4, 7, -3)
		i	their $\overrightarrow{CP}$. their $k(4\mathbf{i} + 8\mathbf{j} - 8\mathbf{k})$ evaluated	M1	allow one numerical error	
		i	$\overrightarrow{AB} \cdot \overrightarrow{CP} = 4 \times (-2) + 2 \times 8 + 1 \times (-8) = 0$ oe	A1	arithmetic must be shown and be consistent <u>Examiner's Comments</u> Most candidates successfully found $\overrightarrow{AB}$ and $\overrightarrow{CP}$ realised the need to find and use a scalar product. Many candidates made sign errors or slipped up with the arithmetic, and gave up. A few ignored the demand to 'show that' and simply stated that the scalar product was zero, which was insufficient.	if 0, allow B4 for fully correct solution using eg Pythagoras or trigonometry
		ii	$\frac{1}{2} \times \text{their } AB \times \text{their } CP$ oe	M1	or $\frac{1}{2}ab\sin C$ from triangle ABC with a , b and C correct	$AB = \sqrt{4^2 + 8^2 + (-8)^2} = 12$ $CP = \sqrt{(-2)^2 + 2^2 + 1^2} = 3$ $AC = \sqrt{5^2 + 4^2 + (-7)^2} = \sqrt{90}$ $BC = \sqrt{1^2 + (-4)^2 + 1^2} = \sqrt{18}$ $AP = 9$ and $BP = 3$ $A = 18.4^\circ$, $B = 45^\circ$ and $C = 116.6^\circ$



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	18	A1	<u>Examiner's Comments</u> Very few candidates made the connection between the work in part (i) and the base and perpendicular height of the required triangle. Most started again, finding an angle using the Cosine Rule or scalar product, and then using $\frac{1}{2}ab\sin C$ – and most lost accuracy somewhere along the way.	
			Total	6		



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Question			Answer/Indicative content	Marks	Part marks and guidance	
23		i	vectors in plane $\begin{pmatrix} 2 \\ 3 \\ 6 \end{pmatrix} \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 5 \end{pmatrix}$ and $\begin{pmatrix} 4 \\ -1 \\ 2 \end{pmatrix} \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ -3 \\ 1 \end{pmatrix}$	M1*	or multiple(s)	or multiple of $\begin{pmatrix} -2 \\ 4 \\ 4 \end{pmatrix}$
		i	$\begin{pmatrix} 1 \\ 1 \\ 5 \end{pmatrix} \times \begin{pmatrix} 3 \\ -3 \\ 1 \end{pmatrix} = \begin{pmatrix} 16 \\ 14 \\ -6 \end{pmatrix} = 2 \begin{pmatrix} 8 \\ 7 \\ -3 \end{pmatrix}$ $\mathbf{r} \cdot \begin{pmatrix} 8 \\ 7 \\ -3 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 8 \\ 7 \\ -3 \end{pmatrix}$	M1 dep*	for M1, method shown or 2 correct elements	
		i		A1		
		i		M1		
		i	$8x + 7y - 3z = 19$	A1	AEF (Cartesian)	
					<u>Examiner's Comments</u> Vector notation was, again this year, generally answered to a good standard. This question was well answered by many candidates. Most candidates tackled the question by finding vectors in the plane and thus finding the cross product of these. Where this approach was followed, virtually the only errors seen were where candidates misread signs in either the question or in their own working. Those who looked to form equations for x, y and z in terms of two parameters sometimes found it difficult to eliminate their parameters.	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$x = -1 + 4\lambda, y = -2 + 3\lambda, z = 6 - 2\lambda$			
		ii	$8(-1 + 4\lambda) + 7(-2 + 3\lambda) - 3(6 - 2\lambda) = 19$	M1		
		ii	$\Rightarrow \lambda = 1$	M1	solves and attempts substitution	
		ii	intersect at (3, 1, 4)	A1	Accept vector form	
					<u>Examiner's Comments</u>	
					Virtually all candidates were aware of the correct methodology for this part. Even those who made an error in (i) usually followed through with the correct method, gaining two marks.	
		iii	$\cos(\alpha) = \frac{\begin{pmatrix} 8 \\ 7 \\ -3 \end{pmatrix} \cdot \begin{pmatrix} 4 \\ 3 \\ -2 \end{pmatrix}}{\sqrt{8^2 + 7^2 + 3^2} \sqrt{4^2 + 3^2 + 2^2}} = \frac{59}{\sqrt{122}\sqrt{29}}$	M1*		can be implied by 7.3° or 0.13 or $\cos \alpha = 0.9919$ seen
		iii	$\theta = \frac{1}{2}\pi - \alpha$	M1dep*	can use $\sin \theta$	consistent use of degrees or radians
		iii	$\theta \approx 1.44$ or $\theta \approx 82.7^\circ$	A1	<u>Examiner's Comments</u>	
					Most successful candidates used the scalar product to solve this problem with roughly equal numbers of those using $\cos \alpha$ to find '90 – required angle' and those directly using $\sin \theta$. A few used the vector product equally effectively. A number of candidates found the angle between / and the normal to the plane but then stopped rather than finding the angle between the plane and the line. A few of them, in error, found the scalar product using the position vector for / rather than its direction vector.	



Question			Answer/Indicative content	Marks	Part marks and guidance
			Total	11	



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Question	Answer/Indicative content	Marks	Part marks and guidance	
24	$\begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix} \times \begin{pmatrix} 4 \\ -2 \\ 3 \end{pmatrix} = \begin{pmatrix} 7 \\ -10 \\ -16 \end{pmatrix}$ $\begin{pmatrix} 3 \\ 1 \\ -1 \end{pmatrix} - \begin{pmatrix} 1 \\ -2 \\ 5 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \\ -6 \end{pmatrix}$ $\text{shortest distance} = \frac{\left \begin{pmatrix} 2 \\ 3 \\ -6 \end{pmatrix} \cdot \begin{pmatrix} 7 \\ -10 \\ -16 \end{pmatrix} \right }{\sqrt{7^2 + 10^2 + 16^2}} = \frac{80}{\sqrt{405}} = \frac{16\sqrt{5}}{9}$ Alternative method after 1st three marks: Forms general vector between lines, equates to $k(7i - 10j - 16k)$ solves to $k = 16/81$ then shortest dist $k 7i - 10j - 16k$ $= \frac{80}{\sqrt{405}} \left(= \frac{16\sqrt{5}}{9} \right)$	M1* M1dep* A1 M1 A1 M1 A1 M1* A1 M1dep*	Direction vectors of lines Vector product Vector between lines Component of their vector in their direction or 3.98	condone 1 error



Question			Answer/Indicative content	Marks	Part marks and guidance	
				A1	or 3.98 <u>Examiner's Comments</u> Solutions for this question were generally very good with most candidates appearing well-versed in applying the formula for the shortest distance between two lines. There were several instances of poor sign transcription. However, it was rare for candidates not to apply a correct method when finding the vector product. One alternate method that candidates applied successfully involved finding a general vector that joined any two points on the lines, then ensuring that it was perpendicular to them by use of scalar products. Another fairly common approach was to convert the problem into finding the shortest distance between two parallel planes each containing one of the lines.	
			Total	7		

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Question			Answer/Indicative content	Marks	Part marks and guidance	
25		i	$\mu = 3 \text{ soi}$	B1		from $3 + 2\lambda = 4 + \mu$ and $5 + \lambda = 10 - \mu$ NB $\lambda = 2$
		i	$1 = 19 + (\text{their } 3) \times \alpha \text{ oe}$	M1		do not allow sign errors
		i	$[\alpha =] - 6$	A1		
		i			<u>Examiner's Comments</u> This was very well-done. The question was accessible to nearly all candidates and many achieved full marks. A few made sign errors or slipped up with arithmetic.	
		ii	$2 \times 1 + 1 \times (-1) + 1 \times \alpha$	M1*	allow 1 sign error	NB $1 + a = \sqrt{6} \times \sqrt{(2 + a^2)} \times \cos 60^\circ$
		ii	$\sqrt{(2^2 + 1^2 + 1^2)} \times \sqrt{(1^2 + (-1)^2 + a^2)} \times \cos 60^\circ$	M1*	allow 1 slip, eg sign error or omission of power	
		ii	eg their $4 + 8\alpha + 4\alpha^2 = 6(2 + \alpha^2)$	M1dep*	square both sides	
		ii	$\alpha = 2 \text{ cao}$	A1	if M1M1M0, B2 for unsupported or alternative valid method	NB $2a^2 - 8\alpha + 8 = 0$
		ii			<u>Examiner's Comments</u> Most candidates adopted the correct strategy and attempted to use the correct scalar product. Many went astray in the arithmetic; it was surprising to see $\sqrt{(1^2 + (-1)^2 + \alpha^2)}$ become $\sqrt{2}\alpha$ so many times. Many candidates attempted to square both sides of the equation, but bracket errors were frequent, so full marks was usually only achieved by the best candidates.	
Total				7		



Question			Answer/Indicative content	Marks	Part marks and guidance	
26		i	$\begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix} \times \begin{pmatrix} 3 \\ 5 \\ 2 \end{pmatrix} = \begin{pmatrix} 7 \\ -7 \\ 7 \end{pmatrix} = 7 \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$	M1		M1 requires evidence of method for cross product or at least 2 correct values calculated
		i		A1		
		i	(eg) $z = 0 \Rightarrow 2x + y = 4, 3x + 5y = 13 \Rightarrow x = 1, y = 2$	M1		or any valid point e.g. (0, 3, -1), (3, 0, 2)
		i	$\mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$	A1	oe vector form	Must have full equation including 'r ='
		i	Alternative: Find one point	M1		
		i	Find a second point and vector between points	M1		
		i	multiple of $\begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$	A1		
		i	$\mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$	A1		
		i	Alternative: Solve simultaneously	M1	to at least expressions for x,y,z parametrically, or two relationship between 2 variables.	
		i		M1		
		i	Point and direction found	A1		
		i	$\mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$	A1		

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Question			Answer/Indicative content	Marks	Part marks and guidance	
		i			Examiner's Comments Vector notation was generally used to a good standard this year. This question was generally answered well with most successful candidates using the vector product approach. Amongst candidates solving simultaneous equations there was some carelessness in arithmetic. There are still candidates who omit "r=" from the vector equation of a line; one suspects that they lack a secure conceptual understanding of the way vector equations operate.	
		ii	$\frac{ 2 \times 2 + 5 - -2 - 4 }{\sqrt{2^2 + 1^2 + 1^2}} = \frac{7}{\sqrt{6}}$	M1	Condone lack of absolute signs for M1	2.86 with no workings scores M1
		ii		A1	oe surd form. isw	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	Alternative: find parameter for perpendicular meets plane and use to find distance	M1	For complete method with calculation errors Examiner's Comments Vector notation was generally used to a good standard this year. A significant number of candidates appear unaware that the Mathematical Formulae booklet (MF1) contains a formula for the perpendicular distance of a point from a plane. Some of those who did not use this formula were able to derive the answer using geometrical reasoning, or an alternative (learnt) standard method. The most common mistake among those who used the given formula was to neglect to consider the equation of the plane with all terms on the left-hand side; this resulted in a wrong method (of adding 4 instead of subtracting it) and consequently scored zero.	look for $\lambda = -7/6$
			Total	6		

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Question			Answer/Indicative content	Marks	Part marks and guidance	
27			$\pm ((3 - 2)i + (-3 - 8)j + (6 - 2)k)$ so i their $\pm (i - 11j + 4k)$. $\pm(5i + 5j + 8k)$ both diagonals used; evaluation not essential $\pm \frac{(1 \times 5 + (-11) \times 5 + 4 \times 8)}{\sqrt{1^2 + 11^2 + 4^2} \times \sqrt{5^2 + 5^2 + 8^2} \cos \theta}$ oe $\theta = \cos^{-1} \frac{\pm 18}{\sqrt{138} \times \sqrt{114}}$ 81.7 to 82°	B1 M1 A1 A1 A1	NB $i - 11j + 4k$ if M0 SC2 for 84° (or 84.5°), or 52(.3°) or 39° or (38.5° or 43(.2°) or 46(.0°) found from scalar product or SC1 for the equivalent obtuse angle must be fully correct 1.4 to 1.43 rad <u>Examiner's Comments</u> Some candidates didn't recognise the term "parallelogram" – the usual mistake was a sketch of a trapezium. Many more didn't seem to recognise the term "diagonal", and successfully used a scalar product to find the acute angle between the sides, or, rarely, one of the other acute angles in OABC. Those who drew a labelled diagram were usually successful. Nearly all candidates were able to demonstrate proficiency with scalar product, and understood "acute".	or B3 for correct use of Cosine Rule (using the midpoint of the diagonals of the parallelogram) $[\cos \theta] = \frac{34.5 + 28.5 - 72}{2\sqrt{34.5}\sqrt{28.5}}$ oe B2 for 81.7 to 82° unsupported or B3 + B2 possible for Cosine Rule
			Total	5		



Question			Answer/Indicative content	Marks	Part marks and guidance	
28		i	$1 - s = 2 + t$ $4 + 2s = 8 + 3t$ $1 + 2s = 2 + 5t$	B1	for all three equations NB third equation may appear later, or with values already substituted	
		i	value of either s or t obtained from valid method	M1		or M1 for one value (of s or t) found from one pair of equations
		i	correct pair of values	A1	eqns (i) and (ii): $s = 0.2, t = -1.2$ eqns (i) and (iii): $s = -\frac{4}{7}, t = -\frac{3}{7}$ eqns (ii) and (iii) $s = 4.25, t = 1.5$	A1 for substitution of this value (of s or t) in third equation and obtaining the other parameter (ie of t or s); NB $(0.2, -0.12)$ or $(-\frac{4}{7}, -\frac{12}{7})$ or $(4.25, -5.25)$ if s found first and $(-2.5, -1.2)$ or $(\frac{19}{14}, -\frac{3}{7})$ or $(-2.5, 1.5)$ if t found first
		i	eg $1 + 2 \times 0.2 \neq 2 + 5 \times -1.2$ oe isw NB A0 for $1 + 2 \times 0.2 = 2 + 5 \times -1.2$ unless clarified by suitable comment	A1	correct substitution of correct values in correct equation <u>Examiner's Comments</u> Most candidates knew what to do here. The majority used the coefficients of i and j to obtain a pair of correct values for s and t , before substituting in a third equation to demonstrate inconsistency. A minority of candidates went astray in the arithmetic and lost one or two of the accuracy marks.	or find same parameter from second pair of equations A1 for correct demonstration of inconsistency NB clear statement needed if two different values of same parameter found
		ii	$2i - 4j - 4k = -2 (-i + 2j + 2k)$ oe	B1	allow equivalent in words, but scale factors must be correct	eg direction of A is $-\frac{1}{2} \times$ direction of C



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	eg line A goes through (1, 4, 1) but line C goes through (1, 15, 11), so they do not coincide so the lines are parallel eg demonstration of different y or z values on each line for (say) $x = 1$, so lines are parallel	B1	<u>Examiner's Comments</u> In part (ii), the majority of candidates understood the relationship, but were unable to convey their understanding well enough to score, often associating the scale factor of -2 with the wrong vector. Very few candidates checked whether the equations represented the same line or two distinct lines. A minority of candidates concluded that the lines were perpendicular.	
			Total	6		



Question			Answer/Indicative content	Marks	Part marks and guidance	
29		i	$l \parallel \begin{pmatrix} 2 \\ 3 \\ 5 \end{pmatrix} \quad \Pi \perp \begin{pmatrix} 4 \\ -1 \\ -1 \end{pmatrix} \quad \text{so} \quad \begin{pmatrix} 2 \\ 3 \\ 5 \end{pmatrix} \cdot \begin{pmatrix} 4 \\ -1 \\ -1 \end{pmatrix} = 0 \Rightarrow l \parallel \Pi$	M1	dot product of correct vectors = 0	Argument can be about a general point on line
		i	$(1, -2, 7)$ on l but $4 \times 1 - -2 - 7 = -1 \neq 8$ so not in Π	M1	substitute point on line into Π and calculate d	
		i	hence l not in Π	A1	Full argument includes key components Examiner's Comments Most candidates scored at least 2 marks, but not all fully justified parallelism; strong answers did this either by referring to perpendicularity to the normal (where they had used dot product) or by saying that the line and plane had no intersection (where they had substituted a general point on the line into the equation of the plane). Some careless arithmetic also resulted in candidates losing the accuracy mark.	
		ii	$(\mathbf{r} =) \begin{pmatrix} 1 \\ -2 \\ 7 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ -1 \\ -1 \end{pmatrix}$	B1		
		ii	closest point where meets Π $4(1 + 4\lambda) - (-2 - \lambda) - (7 - \lambda) = 8$	M1	parametric form of (x, y, z) substituted into plane	
		ii	$\Rightarrow \lambda = \frac{1}{2}$	A1ft		



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$\Rightarrow \mathbf{r} = \begin{pmatrix} 3 \\ -\frac{5}{2} \\ \frac{13}{2} \end{pmatrix}$	A1	Examiner's Comments The initially requested equation of the line was often correctly given although some, erroneously, thought that . In calculating the required point it was common, sadly to see the direction of the line was candidates, using a correct method, deduce that $18t = 9$ but then go on to say that $t = 2$.	
		iii	$\mathbf{r} = \begin{pmatrix} 3 \\ -\frac{5}{2} \\ \frac{13}{2} \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 3 \\ 5 \end{pmatrix}$	B1ft	oe Examiner's Comments This usually followed through correctly from (ii). Again, however, the mark could be lost due to absence of "r = ".	must have "r ="
			Total	8		



Question			Answer/Indicative content	Marks	Part marks and guidance	
30		i	vectors in plane: two of $\begin{pmatrix} -4 \\ 4 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 6 \\ 4 \end{pmatrix} = 2 \begin{pmatrix} 0 \\ 3 \\ 2 \end{pmatrix}, \begin{pmatrix} 4 \\ 2 \\ 3 \end{pmatrix}$	M1	Differences between two pairs	Any multiple
		i	$\mathbf{r} = \begin{pmatrix} 1 \\ 6 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ 3 \\ 2 \end{pmatrix} + \mu \begin{pmatrix} 4 \\ 2 \\ 3 \end{pmatrix}$	A1	Aef of parametric equation Examiner's Comments This question was generally answered well. It was very rare this series to see poor vector notation, which was pleasing. Occasional arithmetic errors were in evidence. Some candidates omitted 'r =' but, surprisingly, the most common error was a transcription error of the sign from candidates' own work when moving from direction vector to equation of line.	Must have "r = ..."
		ii	$\begin{pmatrix} 0 \\ 3 \\ 2 \end{pmatrix} \times \begin{pmatrix} 4 \\ 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 5 \\ 8 \\ -12 \end{pmatrix}$	M1	Calculate vector product	M1 can be awarded where vector product has method shown or only one term wrong
		ii	$\left(\mathbf{r} - \begin{pmatrix} 1 \\ 6 \\ 2 \end{pmatrix} \right) \cdot \begin{pmatrix} 5 \\ 8 \\ -12 \end{pmatrix} = 0$	A1	or multiple	
		ii		M1		
		ii	$5x + 8y - 12z = 29$	A1	Aef of cartesian equation, isw.	Or Cartesian form = d with attempt to compute d
		ii	Alternate method		EITHER	
		ii		M1	x, y, z in parametric form	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii		A1	both parameters in terms of e.g. x, y	
		ii		M1A1	substitute into parametric form of z	
		ii			OR	
		ii		M1	x, y, z in parametric form	
		ii		A1	2 equations in x, y, z and one parameter	
		ii		M1 A1	eliminate parameter	
					Examiner's Comments	
					Those who used a vector product generally had little problem in obtaining the correct equation, with only occasional arithmetic errors. However, those who tried to set up simultaneous equations often were unable to correctly work through the algebra and in many cases gave up part way through their solution. A few candidates lost the last mark by not giving a cartesian form of the equation.	
			Total	6		

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Question			Answer/Indicative content	Marks	Part marks and guidance	
					Examiner's Comments Most candidates were able to form the appropriate equations but too many omitted the first negative sign in the second equation; this was considered to be due to carelessness rather than to misreading. The solution of these equations, with the associated inconsistency, was often not performed systematically but the vast majority, working carefully, fulfilled their objectives. However, one aspect was not clearly shown; examiners were frequently informed that the direction vectors were not equal or were not multiples — but these direction vectors were often not defined.	
			Total	6		

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Question			Answer/Indicative content	Marks	Part marks and guidance	
32		i	$x = 2t + 1, y = 5t - 1, z = t + 2$	B1	Parameterise	or B1 for y and z correctly in terms of x e.g. $2y = 5x - 7$, $2z = x + 3$
		i	$(2t + 1) + 2(5t - 1) - 2(t + 2) = 5$		Substitute into plane	Then M1 for full simultaneous equations method.
		i	$\Rightarrow 10t = 10 \Rightarrow t = 1$	M1	Solve	
		i	Intersect at (3, 4, 3)	A1	cao	Accept vector form
					Examiner's Comments	
					This question was nearly always correctly answered. Where candidates dropped marks it was usually due to basic arithmetic or sign (transcription) errors.	
		ii	$\cos\left(\frac{1}{2}\pi - \theta\right) = \frac{\begin{pmatrix} 2 \\ 5 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix}}{\begin{pmatrix} 2 \\ 5 \\ 1 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix}} = \frac{10}{3\sqrt{30}}$	M1A1		Attempt to find angle or its complement
		ii	$\theta = 0.654$	A1	or 37.5°	
					Examiner's Comments	
					Answers were usually correct, though some candidates gave, as their final answer, the angle between line and normal rather than between line and plane.	
		iii	If P is point of intersection and Q is required point,			
		iii	$\overrightarrow{PQ} = \lambda \begin{pmatrix} 2 \\ 5 \\ 1 \end{pmatrix}$ so $\sin \theta = \frac{2}{PQ} = \frac{2}{ \lambda \sqrt{30}}$	M1*	or $\overrightarrow{PQ} \cdot \hat{\mathbf{n}} = \pm 2$ where $\mathbf{n} = \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix}$	Use $\overrightarrow{PQ}$ with right angled triangle or consider component of $\overrightarrow{PQ}$ in direction of normal vector.



Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$\frac{10}{3\sqrt{30}} = \frac{2}{ \lambda \sqrt{30}}$	M1		Valid method to set up equation in λ alone.
		iii	$\lambda = \pm \frac{3}{5}$	A1		(May work from general point on original equation)
		iii	points have position vectors $\begin{pmatrix} 3 \\ 4 \\ 3 \end{pmatrix} \pm \frac{3}{5} \begin{pmatrix} 2 \\ 5 \\ 1 \end{pmatrix}$	*M1	Dep on 1 st M1	
		iii	points at (1.8, 1, 2.4) and (4.2, 7, 3.6)	A1	cao	
		iii	Alternative: Distance = $\frac{ 2t+1+2(5t-1)-2(t+2)-5 }{\sqrt{1^2+2^2+2^2}} = 2$	M1*		Zero if formula used without substitution in of parametric form.
		iii		A1		
		iii		*M1	Solve	
		iii	$\Rightarrow t = 0.4$ or 1.6	A1	At least one value found	
		iii	(1.8, 1, 2.4) and (4.2, 7, 3.6)	A1		
		iii				
					Examiner's Comments	
					Various different methods were successfully used although those candidates who used the formula (from MF1) were most likely to get a complete solution. A common error was to use $PQ = 2$ rather than $PQ\sin\theta = 2$. Others who used a correct method considered only one possible solution rather than two, often ignoring the modulus symbols which they had earlier used.	
			Total	11		



Question			Answer/Indicative content	Marks	Part marks and guidance	
33		i	In each part, mark the answers, ignoring the labels $AB = \sqrt{91}$; $AC = \sqrt{27}$ or $3\sqrt{3}$ ISW	B1; B1	To invoke MR, evidence must be clear 9.54 or 9.539392..; 5.2(0) or 5.1961524..	171 to 171.317 or 2.99
		i	Attempting to use $\vec{AB} \cdot \vec{AC} = AB \cdot AC \cos \theta$	M1	or $BC^2 = AB^2 + AC^2 - 2AB \cdot AC \cos \theta$	
		i	angle $BAC = 171$ (3 sf) or 2.99 (rad) (3 sf) ISW	A1	Final acute answer [8.68 or 0.152] / choice → A0	
		i			Examiner's Comments In general, the basic work was done well but it was obvious that candidates were more used to finding the angle between two lines (and giving the acute version) than finding a specific angle in a geometrical diagram. In the context of this question there was only one angle, and it happened to be obtuse and it was the obtuse angle only which was accepted. Some of those obtaining it may have been lucky in the directions of the vectors used, though others would have realised the situation. Those using the cosine rule did not get involved with the dual possibility.	
		ii	$6i + 4j - 2k$ or $-6i - 4j + 2k$	B1	seen, irrespective of any labelling	...(-1,-3,-9) or (1,3,9) ... (1,1,5) or (-1,-1,-5)
		ii	$6 \times (-1) + 4 \times (-3) - 2 \times (-9) = 0$ ($\therefore$ perpendicular) AG	B1	oe using (6,4,-2) or (-6,-4,2) and...	
		ii	$6 \times 1 + 4 \times 1 - 2 \times 5 = 0$ ($\therefore$ perpendicular) AG	B1	oe using (6,4,-2) or (-6,-4,2) and...	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii			Examiner's Comments This tended to be universally well done, the only exception lying with those candidates who just said that the scalar products were zero, rather than demonstrating it numerically.	
		iii	(AD =) $\sqrt{56}$ Or $2\sqrt{14}$ or 7.48... soi	B1		
		iii	area $ABC = \frac{1}{2}(\text{their})AB \times (\text{their})AC \times \sin(\text{their})BAC$	M1	($\sqrt{ } = 3.74...$ but M mark, not A)	
		iii	$9.3 \leq V < 9.35$, $9\frac{1}{3}$ ISW	A1	Accept even if (i) angle given as 8.68....	i.e. the acute version not accepted in (i)
		iii			Examiner's Comments Those who had noticed carefully the information obtained from parts (i) and (ii) generally did well in this part. Both the obtuse and the corresponding acute angle were accepted but it was curious to note that either the $\frac{1}{2}$ or the $\sin \theta$ were frequently missing from the area of the base triangle.	
			Total	10		

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Question			Answer/Indicative content	Marks	Part marks and guidance	
34		i	Produce (at least 2) relevant equations	M1	e.g. $1 + 2\lambda = 6 + \mu$, $2 + A = 8 + 4\mu$, $3A = 1 - 5\mu$	Dep previous M1M1A1 earned
		i	Eliminate either λ or μ from 2 of them and solve for the other (μ or λ)	M1	soi by correct (λ , μ)	
		i	$\lambda = 2$ and $\mu = -1$ cao	A1	or e.g. $\lambda = 2$ from 2 different pairs	
		i	Check that $(\lambda, \mu) = (2, -1)$ satisfies all eqns	B1	This must be convincing. Check unusual arguments	
		i	P is (5, 4, 6) cao www	A1	Allow any reasonable vector notation	
					Examiner's Comments	
					Almost everyone knew what to do, but this question prompted a large number of scrappy solutions. Equations were labelled 1, 2 and 3; 2 was doubled and called 3; 1 and 2 were subtracted, μ found and then substituted into... . Which equation was used? Which equation was used for confirming consistency? Candidates should work systematically and state clearly what they are doing.	
	ii		Using $\begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 4 \\ -5 \end{pmatrix}$	M1	i.e. correct parts for direction vectors	Expect $\frac{-9}{\sqrt{14}\sqrt{42}}$
	ii		Using $\cos \theta = \frac{\mathbf{a} \cdot \mathbf{b}}{ \mathbf{a} \mathbf{b} }$ giving value $\frac{n}{\sqrt{a}\sqrt{b}}$	M1	for any 2 meaningful vectors in this question using meaningful scalar product & modulus	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	68.2°... (not 111.8...)	A1	or 1.19 (radians) Examiner's Comments The first question candidates had to ask themselves was "Which aspect of the line equations shall we use?" but only a minority stated the answer. Examiners had to fight their way through the myriad of evaluations to determine what was happening. Fortunately most produced the correct answer, though a not insignificant minority only gave the obtuse angle.	
			Total	8		

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Question			Answer/Indicative content	Marks	Part marks and guidance	
35		i	$AB = \sqrt{(+/-2)^2 + (+/-2)^2 + (+/-4)^2}$	B1	oe	If $AB^2 = AD^2 = 24$, then SR B1 $AB = AD$ to be stated for 2 nd B1
		i	$AD = \sqrt{(+/-2)^2 + (+/-4)^2 + (+/-2)^2}$	B1	oe Examiner's Comments This caused little problem for the majority but there were several who lost marks because they assumed that, seeing the same components (albeit in different positions) for their AB and AD , meant they had no more work to do, as the result was "obvious&&&".	
		ii	midpoint is (3, 5, 0)	B1	Accept any reasonable vector notation.	
		ii	Clear method for finding direction vector	M1	Expect $3j - k$ or $-3j + k$	
		ii	$r = 3i + 2j + k + \lambda(3j - k)$ oe or e.g. $r = 3i + 5j + \mu(-3j + k)$ cao	A1	"r =" is essential. No ft. for wrong mid-point. Examiner's Comments Those who found the mid-point correctly usually went on to find the equation correctly, and many who found an incorrect mid-point often knew what to do to obtain the direction vector.	
		iii	substitution of $\lambda = +/- 5$ or $\mu = +/- 4$	M1	Based on correct answer to (ii) Examiner's Comments This part was straightforward for those who had found the equation of the line correctly.	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		iv	Kite	B1	Examiner's Comments The correct answer was the least common of all the answers suggested.	
			Total	7		



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Question			Answer/Indicative content	Marks	Part marks and guidance	
36		i	$\begin{pmatrix} 1 \\ 2 \\ 5 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$	M1		
		i	$\cos \theta = \frac{15}{\sqrt{1^2 + 2^2 + 5^2} \sqrt{2^2 + (-1)^2 + 3^2}} = \frac{15}{\sqrt{30}\sqrt{14}}$	A1	Accept unsimplified	
		i	$\theta = 0.750$ or 43.0°	A1	If zero, then sc1 for $n_1 \cdot n_2 = 15$ seen <u>Examiner's Comments</u> This first question should have been straightforward for any properly prepared candidate since both parts are listed as techniques required in the vectors section of the specification. So it was surprising how many candidates did not answer this with ease. Most candidates used the scalar product method to find an angle, although one or two, perfectly acceptably, chose to use the cross product. It was disappointing to find that some candidates thought that $ a b \sin \theta$ was the value of the scalar product. A few were unfamiliar with the description 'acute angle' or did not know the relationship between the angle between planes and the angle between their normals.	
		ii	$\begin{pmatrix} 1 \\ 2 \\ 5 \end{pmatrix} \times \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} = \begin{pmatrix} 11 \\ 7 \\ -5 \end{pmatrix}$	M1		M1 requires evidence of method for cross product or at least 2 correct values calculated
		ii		A1		
		ii	(eg) $x = 0 \Rightarrow 2y + 5z = 12, -y + 3z = 5 \Rightarrow y = 1, z = 2$	M1		or any valid point e.g. $(-11/7, 0, 19/7)$ $(22/5, 19/5, 0)$



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$\mathbf{r} = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 11 \\ 7 \\ -5 \end{pmatrix}$	A1	oe vector form <u>Examiner's Comments</u> This first question should have been straightforward for any properly prepared candidate since both parts are listed as techniques required in the vectors section of the specification. So it was surprising how many candidates did not answer this with ease. Most candidates knew that they needed to find the direction of the line by using the vector product of the normals to the planes; a few made errors in actually finding it. Finding a point on the line proved more of a challenge for weaker candidates with some fixing values for two of the variables before trying to find the third. Those candidates who attempted to find the line solely through simultaneous equations were far more likely to introduce errors.	Must have full equation including 'r ='
		ii	Alternative: Find one point	M1		
		ii	Find a second point and vector between points	M1		
		ii	multiple of $\begin{pmatrix} 11 \\ 7 \\ -5 \end{pmatrix}$	A1		
		ii	$\mathbf{r} = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 11 \\ 7 \\ -5 \end{pmatrix}$	A1		
		ii	Alternative: Solve simultaneously	M1	to at least expressions for x,y,z parametrically, or two relationship	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	Point found	A1	between 2 variables.	
		ii	Direction found	A1		
		ii	$\mathbf{r} = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 11 \\ 7 \\ -5 \end{pmatrix}$	A1		
			Total	7		



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Question			Answer/Indicative content	Marks	Part marks and guidance	
37		i	$\begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix} \times \begin{pmatrix} 4 \\ -1 \\ -1 \end{pmatrix} = \begin{pmatrix} -4 \\ -2 \\ -14 \end{pmatrix} = -2 \begin{pmatrix} 2 \\ 1 \\ 7 \end{pmatrix}$	M1		
		i		A1	Or any multiple	
		i	$\begin{pmatrix} 3 \\ 0 \\ 1 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ -2 \\ 0 \end{pmatrix}$	B1	Or negative	Or use of $n.(a_1 + pb_1 + kn) = n.(a_2 + qb_2)$ B1 followed by attempt to calculate magnitude of kn M1
		i	shortest distance = $\frac{\left \begin{pmatrix} 2 \\ -2 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 1 \\ 7 \end{pmatrix} \right }{\sqrt{2^2 + 1^2 + 7^2}} = \frac{2}{\sqrt{54}}$ oe	M1	Component of their vector in their direction	
		i		A1	<u>Examiner's Comments</u> Many candidates knew a standard formula for this request and most were then able to use that formula successfully. Others found success by working from first principles; once they had compared the vector between two general points with the perpendicular to both lines, these candidates either used the dot product to simplify or solved the three equations in three unknowns. Marks were sometimes lost through sign errors, with some candidates being unaware of the sign for the j component of the vector product.	
		ii	$2x + y + 7z = \dots$	B1ft	ft from 4(i) only if 1 st M1 mark gained	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	...11	B1 dep	If zero, then sc 1 for any correct vector equation. <u>Examiner's Comments</u> Finding the cartesian equation of the plane was generally answered well by candidates who had found the direction of the perpendicular in part (i). Some, however, left their answer in vector equation form.	
			Total	7		



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